
Do LLM-Judge Biases Compose? A Label-Free Test for the Position $\times$ Verbosity Interaction

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Abstract

Whether an LLM judge yields a valid evaluation depends not only on its accuracy but on how its error profile composes with the debiasing pipeline wrapped around it and with the training or ranking decisions its scores feed. We study one such composition: pipelines correct position and verbosity bias *sequentially*, assuming the two add. We show this assumption is (a) consequential upstream and downstream, (b) checkable without human labels, and (c) currently unchecked. **Consequential:** if position and verbosity bias interact ($\beta_{PV} \neq 0$), a pipeline that judges each pair once and *subtracts* a fitted position term leaves a residual of $\frac{1}{2}|\beta_{PV}|$ in the doubly-biased cell that no ordering removes (Prop. 1); a pipeline that *averages* over both orders removes it on the log-odds scale (up to a Jensen gap), so which deployed designs are exposed is nameable (§3), and a residual of this size shifts leaderboard order and a measurable fraction of RLHF preference labels (§4). **Checkable:** the **Swap-Symmetry Interaction Test** (SSIT) is a difference-in-differences over the four position $\times$ verbosity swap cells; its readout $\hat{i}_{PV}$ is a difference of four cell means, so it estimates the population interaction in win-rate points with *no human labels and no model of the judge* (recovered to 0.004 across six decision rules including a linkless one; Prop. 2). It is not free—it needs a dedicated collection ($\approx 8 \times$ the judgment cost of one evaluation plus a length manipulation)—but it needs no preference labels, which is what an automatic judge is meant to avoid. We give a power analysis for sizing it, a decision procedure, and a checklist framed as input to a judge-deployment disclosure. For $1 \times$ -budget pipelines with no labels, **Interaction-Aware Swap Correction** (IASC) shrinks each position-coupled bias term toward zero (Prop. 3)—a floor, not a fix. **Unchecked:** a pilot on seven open judges over six families (≈ 20 pairs each, no labels) finds $\hat{i}_{PV} > 0$ on six (+0.07 to +0.17; wild cluster bootstrap $p \leq 0.03$), AIA-significant on four; the seventh picks the stronger response on every pair, so SSIT correctly returns 0 (the “too-easy stimulus” regime). For four re-collected judges, a length-only *padding* manipulation keeps the pooled interaction (at half the magnitude) but splits by judge—it grows for one, holds for two, and vanishes for one whose $P \times V$ was position $\times$ style—so a paraphrase probe alone can mislead about the mechanism. Code, data, and a valid-CI (PPI) variant released.

1 Introduction

LLM-as-a-judge evaluation [1, 2] supplies reward signals for RLHF [15] and determines leaderboard rankings [9]. Three systematic biases are well documented—positional ($\mathcal{B}_P$), verbosity ($\mathcal{B}_V$), self-preference ($\mathcal{B}_S$)—each with a published mitigation [1, 9, 7]. Deployed pipelines apply them one after another: swap positions and average, then length-control, then diversify the panel.

The gap. No prior work asks whether this chaining is *valid*—whether the biases compose additively—and the natural way to check, measuring the interaction against human preference labels, defeats the purpose of an automatic judge. The validity question here is not “how accurate is the judge” but “how does its error compose with the pipeline”: the same judge can be well-calibrated in isolation and leave a systematic residual once its position and verbosity corrections are chained. This paper is a practical treatment of the check when there are no labels:

1. **Which pipelines are exposed** (§3, Prop. 1, Cor. 1). This is a direct answer to a standing question—*does position-swap averaging fully debias pairwise evaluation?* A $1\times$ pipeline that fits and subtracts a position term leaves $\frac{1}{2}|\beta_{PV}|$; a $2\times$ swap-averaging pipeline cancels β_P and β_{PV} on the log-odds scale (both odd in P), leaving only a probability-scale Jensen term; non-position pairs ($V\times S, \dots$) it does not touch. Joint estimation is exact.
2. **Downstream cost** (§4): an uncorrected interaction of the win-rate-scale magnitude our worked example suggests reorders adjacent models on a leaderboard and mislabels an estimable fraction of length-mismatched RLHF comparisons.
3. **The check** (SSIT, §5, Prop. 2): a difference-in-differences over the four $P\times V$ swap cells. Its readout $\hat{i}_{PV}$ is a difference of four sample means, hence *design-based*—no human labels, no model of how the judge decides (§5.1: recovered to 0.004 across six decision rules, one with no link), at a dedicated collection cost of $\approx 8\times$ one evaluation (§5). A power analysis sizes it (§5.2).
4. **A $1\times$ -budget correction** (IASC, §6, Prop. 3): shrinks each position-coupled bias term toward zero without labels; a floor, not a fix.
5. **A seven-judge pilot, a decision procedure, and a deployment-disclosure checklist** (§7–8): $\hat{i}_{PV} > 0$ on six of seven open judges across six families, correctly null on the seventh; a length-only re-test of four judges shows the interaction persists pooled but its mechanism (length vs. paraphrase-style) is judge-specific; plus a pre-registered full study.

A valid-CI version via prediction-powered inference [19, 20] is in Appendix C. Self-preference and other factor pairs use the same contrast (Appendix F).

Scope. The only non-simulation numbers are the propositions and the worked example (§7), which demonstrates the diagnostic, not judge behaviour. Simulations use *stipulated* interaction magnitudes to check estimator recovery and power and to bound the downstream cost, not to size the effect for real judges—that is what the pre-registered study (§8) is for.

2 Related Work

Single biases, measured in isolation. Position, verbosity, and self-preference bias are each well characterised with a published mitigation [1, 5, 4, 6, 9, 7, 8]; taxonomies and audits [10, 2, 3, 11] keep the one-bias-at-a-time convention. Length-controlled AlpacaEval [9] removes verbosity as a *marginal* confounder by regression—the regime where Prop. 1 bites.

Multi-bias measurement. The closest empirical work [12] evaluates nine debiasing strategies over five judges and four bias types, and reports two facts we build on: *style* bias (markdown vs. plain prose) dominates—up to 0.76—while position bias *in isolation* is small (≤ 0.04), and verbosity bias sign-flips by family (concise-preferring for Claude, length-preferring for the rest), which our pilot reproduces. It measures each bias type against human labels on a fixed panel; it does not ask whether the corrections *compose*. A small main effect can still carry a consequential interaction—and our worked example measures $P\times(\text{paraphrase})$, which overlaps that dominant style axis [16]. Our contribution is orthogonal: a *label-free* test a practitioner runs *per judge* at deployment scale, where gold labels are what is missing. Per-instance detectors [13] act on several biases at once but give no population-level guarantee.

Prediction-powered inference for evaluation. PPI/PPI++ [17, 18] combine a small gold set with a large model-labelled set for valid CIs; [19] brings this to LLM-based ranking evaluation, and [20] relates PPI-style estimators to the efficient influence function and characterises when they beat measurement-error corrections. All learn a rectifier $\mu(\hat{Y})$ of the judge score alone; Prop. 4 shows that is insufficient under interaction-induced cell heterogeneity, and a factorial rectifier $\mu(\hat{Y}, P, V, PV)$ restores validity (Appendix C).

3 Sequential Correction Leaves a Residual

Model the judge’s choice as logistic in binary bias indicators—position P , verbosity V , self-preference S —and a quality term:

$$\text{logit}P(Y=A) = \gamma(Q) + \beta_P P + \beta_V V + \beta_S S + \beta_{PV} PV + \beta_{PS} PS + \beta_{VS} VS + \beta_{PVS} PVS. \quad (1)$$

Sequential correction (SC) assumes *additive independence* (AIA): all interaction terms are zero. It is a two-stage pipeline: estimate $\hat{\beta}_P$ by marginal regression of Y on P and subtract $\hat{\beta}_P P$; then estimate $\hat{\beta}_V$ from position-matched data (in swap pipelines, the reference stratum $P=0$) and subtract $\hat{\beta}_V V$.

Proposition 1 (SC residual under interaction). *Under (1) restricted to P, V , balanced ($P \perp V$, $\Pr = \frac{1}{2}$), $\beta_{PV} \neq 0$: $\hat{\beta}_P^{\text{marg}} \xrightarrow{P} \beta_P + \frac{1}{2}\beta_{PV}$ (omitted-variable bias), $\hat{\beta}_V \xrightarrow{P} \beta_V$, and after both corrections the $(P=1, V=1)$ cell retains a residual log-odds $\Delta_{\text{SC}} = (\beta_P + \beta_V + \beta_{PV}) - (\beta_P + \frac{1}{2}\beta_{PV}) - \beta_V = \frac{1}{2}\beta_{PV}$. Any symmetric variant of the pipeline leaves an uncorrected term of order $\frac{1}{2}|\beta_{PV}|$ somewhere in the 2×2 table.*

Corollary 1 (Does swap-averaging fully debias? Which pipelines are exposed). *The residual depends on how position bias is removed. (i) **Subtract-a-fitted-term** ($1 \times$ judging, then remove $\hat{\beta}_P P$ from the score, AlpacaEval-style regression debiasing): exposed, residual $\frac{1}{2}\beta_{PV}$. (ii) **Swap-average** (judge both orders, average the score): the $(2P-1)$ -odd terms— β_P and $\beta_{PV}(2P-1)V$ —cancel in the log-odds. Pipelines average the probability/score, and $\frac{1}{2}[\sigma(x) + \sigma(y)] \neq \sigma(\frac{1}{2}(x+y))$, so a second-order Jensen term of size $O(\beta_{PV} \cdot |\delta + \beta_V V|)$ survives; it is small when the two orders have similar baseline rates and 0 at $\beta_{PV}=0$. $V \times S$ and other non-position pairs are untouched by swap-averaging. (iii) **Joint estimation** ($P+V+PV$ regression): exact. Case (i) needs SSIT to decide whether to act; case (ii) needs it for the non-position pairs and, if the baseline-rate gap is large, for $P \times V$ too.*

The algebra is textbook omitted-variable bias [22]; the contribution is pinning the exposure to a design—the answer to “does swap-averaging debias” is *yes for the position $\times$ verbosity interaction on the log-odds scale, no for the surface-form and self-preference pairs, and not at all for the widely-used subtract-a-term pipelines*. To know whether it bites for your judge you must measure β_{PV} —§5.

4 Downstream: What an Uncorrected Interaction Costs

Write ι_{PV} for the interaction on the *win-rate* scale—the probability-scale image of the log-odds β_{PV} from §3, and the quantity a debiased pipeline should drive to zero (§5 estimates it without labels). An uncorrected ι_{PV} biases the measured win rate on length-mismatched comparisons by up to ι_{PV} win-rate points, and that bias flows into whatever those win rates feed (`sim_ranking_impact.py`). We take $\iota_{PV}=0.15$ —within the $+0.07$ to $+0.17$ range our pilot shows (§7)—as a running figure.

Leaderboards. In a constructed six-model leaderboard on a tight skill ladder (adjacent true win rate $\approx 51\%$), a judge carrying $\iota_{PV}=0.15$ inverts the ranking under no correction and under sequential correction alike (Kendall τ to truth = $+0.20$; worst model above best), because the differential lives entirely in β_{PV} . Log-odds swap-averaging and joint estimation restore $\tau=1.0$; label-free IASC recovers $\tau=0.87$. The construction is adversarial (real leaderboards randomise position), so the absolute drop is an upper bound—but the ordering of *corrections* is the point, and it is stable across a $4 \times$ sweep of the bias magnitude (Appendix G).

Preference data. On the affected cell the judge’s pick-target probability is shifted by up to ι_{PV} relative to the quality-only preference. As a loose upper bound on label-level impact: if quality preferences cluster near 50/50 this sets the labelled winner by the interaction rather than quality for 26–44% of that cell at $\iota_{PV}=0.15$ (9–26% for more decisive preferences, $\iota_{PV}=0.05$ – 0.15). For a preference corpus with a length gap on a fraction f of comparisons, halve those for the corpus-level rate. These labels then compound into model weights through RLHF/DPO [15], and—because verbose responses are systematically favoured on the very cell where position bias is cancelled—in a direction that rewards verbosity.

5 A Label-Free Test for the Interaction

Run every pair through all four (P, V) swap cells. Swapping the two responses maps $P \rightarrow 1-P$ and (pick-target) $\rightarrow$ (pick-other), so the judge’s dependence on position must be *odd* in P : write it

$\beta_P(2P-1)$, with the interaction $\beta_{PV}(2P-1)V$. The pooled log-odds of picking the target is then

$$\text{logit}\pi_1(P, V) = \delta + \beta_P(2P-1) + \beta_V V + \beta_{PV}(2P-1)V, \quad (2)$$

$\pi_1 = \Pr(\text{pick target})$, δ the unknown per-pair quality gap.

Proposition 2 (Swap-symmetry identity). *The difference-in-differences of the four cell log-odds, $\Delta_{SS} = [\text{logit}\pi_1(1, 1) - \text{logit}\pi_1(0, 1)] - [\text{logit}\pi_1(1, 0) - \text{logit}\pi_1(0, 0)]$, equals $2\beta_{PV}$ and does not depend on δ , β_P , or β_V (Appendix B).*

We call the plug-in the **Swap-Symmetry Interaction Test** (SSIT). It has two readouts, and it matters which you trust:

- $\hat{i}_{PV}$ — the **same DiD on the probability scale**: a difference of the four within-pair cell means, $[\bar{r}_{11} - \bar{r}_{01}] - [\bar{r}_{10} - \bar{r}_{00}]$. This is *design-based*: it estimates the population interaction in win-rate points under *any* judge decision rule—no link, no additivity, no homogeneity. (2) is not needed for it.
- $\hat{\beta}_{PV}$ — the log-odds coefficient from the saturated 8-cell fit. This *does* use (2) and attenuates when pooling over heterogeneous δ ; report it only as a rough secondary number, and only with ≥ 3 samples/cell.

No gold labels enter either; a small gold set is needed only to put a CI on the debiased win rate itself (Appendix C). The same contrast on the other factor pairs gives ι_{PS}, ι_{VS} .

Cost. SSIT is label-free but not free. It needs each calibration pair judged in all four (P, V) cells at ≥ 2 samples— $\approx 8\times$ the judgment cost of scoring that pair once—plus a verbosity manipulation (a longer version of one response) that a debiasing pipeline does not otherwise produce. It is a dedicated experiment on a calibration subset, cheaper than collecting human preferences but not a free rider on the two-order swap that debiasing already runs. The trade is: no preference labels, which is the point of an automatic judge.

5.1 What does $\hat{i}_{PV}$ actually assume?

Almost nothing. We plant a judge, compute *that judge's* true population ι_{PV} by high- N Monte Carlo, and check whether SSIT recovers it from 60-pair panels with no labels, across six decision rules—including one *out of family*: heavy-tailed (Student- t) quality perception with a hard confidence threshold and no link function at all (`sim_ssit_robustness.py`).

Table 1: SSIT's $\hat{i}_{PV}$ vs. the judge's *true* population ι_{PV} , no gold labels, 60 pairs $\times$ 300 panels. “bias” = mean $\hat{i}_{PV}$ minus the truth for that DGP. The asymmetric row has a smaller true ι_{PV} because that judge is genuinely unbiased on easy pairs—SSIT reports it correctly, it is not attenuated.

Judge decision rule	true ι_{PV}	SSIT $\hat{i}_{PV}$	bias	$\Pr(\hat{i}_{PV} > 0)$
additive logistic (SSIT's own model)	+0.24	+0.24 (± 0.08)	−0.00	1.00
logistic, wide quality-gap spread	+0.20	+0.20 (± 0.07)	−0.00	1.00
bias slopes vary with quality, symmetric	+0.22	+0.22 (± 0.08)	+0.00	1.00
bias fades when target clearly better	+0.16	+0.16 (± 0.08)	−0.00	0.98
probit link (SSIT assumes logit)	+0.33	+0.33 (± 0.07)	−0.00	1.00
out of family : t -noise + threshold, no link	+0.23	+0.23 (± 0.07)	−0.00	0.99

Bias ≤ 0.004 everywhere, including the linkless judge—because $\hat{i}_{PV}$ is a difference of four sample means and estimates its population counterpart whatever the judge does. (2) only buys the interpretation as a log-odds coefficient. The one thing $\hat{i}_{PV}$ relies on is the *design*: the same pairs in all four cells, judged independently (no cross-cell memory effects); the pre-registered protocol randomises cell order to protect this. $\hat{i}_{PV}$ is the average win-rate distortion over the pair population tested—not the mechanism, and not a statement about any other pair mix.

5.2 How large a check do you need?

$\hat{i}_{PV} = \bar{d} = \frac{1}{n} \sum_k d_k$ is a sample mean of the per-pair DiDs d_k , so by the CLT $\bar{d} \approx \mathcal{N}(\iota_{PV}, \sigma_d^2/n)$ with $\sigma_d^2 = \text{Var}(d_k)$, and the two-sided level- α wild-cluster test has power

$$\text{power} \approx \Phi\left(\frac{|\iota_{PV}|\sqrt{n}}{\sigma_d} - z_{1-\alpha/2}\right), \quad n_{80\%} \approx \frac{7.8 \sigma_d^2}{\iota_{PV}^2}. \quad (3)$$

This is *a priori*: it needs only σ_d , which splits into a between-pair term (heterogeneous δ_k , judge stochasticity) and a sampling term $\propto 1/s$ for s samples/cell, and which you can pin from a ≤ 10 -pair pre-run or borrow from a similar judge. A decisive judge ($\sigma_d \approx 0.25$, as in §7) needs $n \gtrsim 22$ for a 0.15 effect; a mid-range one ($\sigma_d \approx 0.4$) needs $n \gtrsim 55$. Table 2 is the wild-bootstrap simulation of the same quantity—it tracks (3) and is mildly conservative below $n=30$ (the bootstrap estimates σ_d rather than knowing it).

Table 2: Power of the SSIT wild-cluster test (5%), 2 samples/cell, mid-range judge (cell rate ≈ 0.8), 500 panels. Near-deterministic judges reach these rows at smaller n ; 1 sample/cell roughly halves each entry (`sim_ssit_power.py`).

true ι_{PV}	20 pairs	30 pairs	50 pairs	100 pairs
0.05	0.08	0.11	0.11	0.16
0.10	0.14	0.20	0.26	0.47
0.15	0.25	0.35	0.52	0.76
0.20	0.38	0.54	0.75	0.95

Rule of thumb (from (3)): budget $n \approx 8 (\sigma_d/\iota_{PV})^2$ pairs at ≥ 2 samples/cell— ≈ 25 for a decisive judge and a 0.15 target, ≈ 55 for a mid-range one. After running, substitute the observed SD(d_k) (the realized-power column of Table 3).

6 IASC: A Partial Label-Free Correction

The swap contrasts that identify β_{PV} also identify β_P (and β_{PS}, β_{PVS} with a third factor): every coefficient whose monomial carries the $(2P-1)$ factor, i.e. the *position-coupled subspace*. Reading these off a swap-augmented calibration subset and subtracting them gives **Interaction-Aware Swap Correction**:

$$\hat{s}_i = \sigma(\text{logit} \hat{Y}_i - \hat{\beta}_P(2P_i-1) - \hat{\beta}_{PV}(2P_i-1)V_i). \quad (4)$$

Proposition 3 (IASC shrinks each position-coupled term). *Fitting a common slope across pairs with heterogeneous δ_k attenuates each swap-cell coefficient, $\hat{\beta}_P \xrightarrow{P} \lambda_P \beta_P$ and $\hat{\beta}_{PV} \xrightarrow{P} \lambda_{PV} \beta_{PV}$ with $\lambda_P, \lambda_{PV} \in (0, 1]$ increasing to 1 as $\text{Var}(\delta_k) \rightarrow 0$ (Appendix B). Then (4) gives $\mathbb{E}[\text{logit} \hat{s}_i | k] = \delta_k + (1-\lambda_P)\beta_P(2P_i-1) + \beta_V V_i + \beta_S S_i + (1-\lambda_{PV})\beta_{PV}(2P_i-1)V_i$: a positive fraction of each position-coupled term is removed, sign preserved, and β_V, β_S untouched.*

Two honest caveats. (i) λ_P and λ_{PV} need not be equal, so a cell can retain oppositely-signed main-effect and interaction residuals. (ii) The guarantee is on the *log-odds*; after the $\sigma(\cdot)$ the corrected pick rate need not lie between the raw and swap-averaged rates—on the pilot (§7) it moves $0.93 \rightarrow 0.98$ past the $2\times$ swap-average of 0.95 (Table 4), a Jensen overshoot. IASC is a heuristic floor for the log-odds bias, not a guaranteed improvement on the win rate.

When to use it. IASC fits one situation: you judge each pair *once*, have *no* gold labels, and want back part of what $2\times$ swap-averaging would remove. It removes ≈ 60 – 80% of each position-coupled log-odds term on quality-matched calibration pairs, less otherwise (Table 1), for one extra swapped pass on a calibration subset. With a gold set, use the factorial-rectifier estimator instead (Prop. 4, Appendix C).

7 Pilot: Seven Judges, Six Families

We ran SSIT gold-free on seven open judges spanning six model families—gpt-oss-20B/120B (Groq), gemma (Google AI Studio), qwen3-27b (Groq qwen/qwen3.8-27b), allam-2-7b (Groq), mistral-small, command-r (Cohere)— ≈ 20 pairs each, 1–2 samples/cell (Table 3). $n \approx 20$ is below what §5.2 recommends for a mid-range judge, and V is a paraphrase (below), so we read this as a multi-judge feasibility test, not a magnitude estimate.

Table 3: SSIT on seven judges, ≈ 20 pairs/judge. $\hat{i}_{PV}$ = probability-scale swap-cell DiD; “wild CI” / p = wild cluster bootstrap (Rademacher, $B=9,999$); “rlz. pow.” = design-based realized power from the observed $SD(d_k)$ (`analyze_real.py`); “AIA” = both the $\hat{i}_{PV}$ and log-odds $\hat{\beta}_{PV}$ CIs exclude 0.

Judge	family	$\hat{i}_{PV}$	wild 95% CI	wild p	rlz. pow.	AIA	$\hat{b}_V$
<code>gpt-oss-20b</code>	openai	+0.12	[+0.01, +0.21]	0.026	0.64	yes	+0.5
<code>gpt-oss-120b</code>	openai	+0.17	[+0.04, +0.30]	0.009	0.69	yes	+1.5
<code>gemma</code>	google	+0.16	[+0.07, +0.26]	0.0001	0.91	yes	−1.4
<code>qwen3-27b</code>	qwen	+0.12	[+0.05, +0.20]	0.0008	0.86	yes	+0.8
<code>allam-2-7b</code>	allam	+0.09	[+0.01, +0.17]	0.034	0.61	no	+0.8
<code>mistral-small</code>	mistral	+0.07	[+0.01, +0.14]	0.032	0.48	no	+1.4
<code>command-r</code>	cohere	+0.00	[+0.00, +0.00]	1.0	—	no	+0.0
Pooled (7)	—	+0.10	[+0.05, +0.14]	≈ 0	0.98	yes	—
Pooled (excl. <code>command-r</code>)	—	+0.12	[+0.06, +0.17]	≈ 0	—	yes	—

Six of seven show it; the seventh is unchallenged. $\hat{i}_{PV}$ is positive on every judge except `command-r` (+0.07 to +0.17), AIA-significant on four judges across three families (`gpt-oss-20B/120B`, `gemma`, `qwen3-27b`; wild $p \leq 0.026$, both bootstrap CIs excluding 0), and positive-but-marginal on `mistral-small` and `allam-2-7b` (wild $p \approx 0.033$, pairs-bootstrap $\hat{i}_{PV}$ CI touching 0; neither survives a Bonferroni correction over the three factor pairs). The detecting judges have $SD(d_k) \approx 0.2$ – 0.3 , so by (3) $n \approx 20$ affords 60–90% power for a 0.15 effect—the detections are what the a-priori formula predicts, not luck. `command-r` picks the stronger response on all 20 pairs, so $\hat{i}_{PV}=0$ exactly—not because it is unbiased but because our r_2 responses are visibly weaker, the regime where Table 1’s “bias fades when the target is clearly better” row applies and where SSIT correctly returns null. It is why the pre-registered design uses quality-matched pairs. The $\hat{b}_V$ sign flip persists (concise-preferring `gemma` vs. length-preferring rest), so the interaction is not one model’s length taste.

A length-only probe separates position $\times$ length from position $\times$ style. Our default V rewrites r_1 into a longer form, so it entangles length with rephrasing—and [12] reports that *style* bias, not position bias, dominates. We re-collect four judges with a *padding-only* V that keeps r_1 verbatim and appends redundant restatement of points already made (`make_padded_v.py`), so V toggles length alone. Pooled, the interaction persists ($\hat{i}_{PV}=+0.06$ excl. `command-r`, wild $p=0.003$), about half the paraphrase- V magnitude—but the mechanism is judge-specific: `mistral-small` *grows* (+0.075 $\rightarrow$ +0.10, wild $p=0.003$), `allam-2-7b` holds marginally (+0.09 $\rightarrow$ +0.075, $p=0.07$), `command-r` stays 0, and `qwen3-27b`’s interaction *vanishes* (+0.12, $p=0.0008 \rightarrow -0.02$, CI [−0.18, +0.15], $p=0.85$): for that judge the paraphrase-driven component carried $P \times V$ and the length-only signal is null (with the higher verdict-drop caveat of Appendix D). A paraphrase V and a padding-only V give opposite answers for the same judge—which is exactly why the pre-registered protocol (Appendix A) mandates a length-only manipulation.

What it does not establish. An effect size for a deployed judge: $n \approx 20$, no frontier judge, and the padding-only re-test covers four of seven. [12]’s ≤ 0.04 position bias is on the *main effect*; our $\hat{i}_{PV}$ is the interaction, which the residual argument (§3) says a sequential pipeline leaves behind *whether the non-position factor is length or style*—the proof does not care about the mechanism. Self-preference and $V \times S$ (Appendix F) are weaker and, for S , confounded. $\approx 15\%$ of `gemma` verdicts are dropped (Appendix D).

8 Discussion

The additivity assumption behind chained bias correction is now cheap to check and cheap to act on. We close with the operational version.

Decision procedure. For an evaluation that reports a sequentially position-corrected win rate:

1. If the pipeline *swap-averages* the log-odds and the two orders have similar baseline rates: $P \times V$ is handled (Cor. 1); run SSIT only for the non-position pairs.

2. Else, run SSIT per judge on enough pairs for $\geq 80\%$ power at your smallest effect of concern (Table 2: ≈ 50 –100 at 2 samples/cell for a 0.15 effect, fewer for a decisive judge); report $\hat{i}_{PV}$ with a wild-cluster CI *and* the realized power from the observed $SD(d_k)$.
3. CI excludes 0 $\Rightarrow$ act (step 4). CI contains 0 *and* realized power $\geq 0.8 \Rightarrow$ sequential correction is defensible for effects above your threshold. CI contains 0 *and* power $< 0.8 \Rightarrow$ **inconclusive**: collect more pairs/samples, do not treat as a pass.
4. Act: switch to log-odds swap-averaging or joint $P+V+PV$ estimation; if stuck at $1\times$ with a gold set, use the factorial-rectifier estimator (Appendix C); if $1\times$ and no labels, apply IASC as a floor.

Deployment-disclosure checklist. A judge that gates training or ranking decisions should ship with an evaluation-pipeline disclosure—the analog of a model card for the evaluator—and the $P\times V$ interaction is one line of it. Disclose: (1) how position bias is removed (subtractive / log-odds-averaged / joint); (2) $\hat{i}_{PV}$ (and the relevant non-position pairs) with a wild-cluster CI and the realized power; (3) whether the verbosity probe was length-only or a paraphrase; (4) what the judge’s scores feed upstream/downstream. The same four questions are what a *reviewer* of a judge-based evaluation should ask.

Pre-registered full study (Appendix A). ≥ 8 judges over ≥ 5 families *including frontier deployed judges*, quality-matched stimuli, ≥ 50 pairs $\times$ 4 tasks, all eight (P, V, S) cells, ≥ 3 samples/cell, a *padding-only* verbosity manipulation, a held-out generator for S , wild cluster bootstrap, Bonferroni over the three pairs.

Limitations. Every quantitative magnitude outside the propositions and the pilot $\hat{i}_{PV}$ is simulation-based with stipulated interaction terms—illustrative. The pilot is seven small open judges, ≈ 20 pairs each (below the recommended n), no frontier judge; one judge’s stimulus was too easy to register any bias; the paraphrase-vs-padding check covers four judges (§7). IASC and IA-PPI are post-hoc additions to the pre-registered endpoint; IA-PPI is post-stratified PPI, not a new inference principle. SSIT is a difference-in-differences—the contribution is that swap symmetry makes it label-free and per-judge, plus the exposure taxonomy (Cor. 1), the power analysis, and the decision procedure, not the estimator.

Reproducibility Statement

Seed 2024 throughout. Released (numpy/scipy only): `ssit.py` (SSIT, pairs + wild cluster bootstrap, gold-free self-test), `iasc.py` (IASC, Eq. 4), `ia_ppi.py` (IA-PPI reference impl.), `sim_prop2_coverage.py` (Appendix C), `sim_ssit_robustness.py` (Table 1), `sim_ssit_power.py` (Table 2), `sim_ranking_impact.py` (§4: leaderboard τ , RLHF label-flip bound, and Appendix G), `analyze_real.py` (regenerates Table 3, realized power, and the three-pair / by-task breakdown of Appendix F from the raw pilot judgments in `data/raw/`; `data/raw_padded/` + `make_padded_v.py` for the length-only re-test), `analyze_iasc.py` (IASC on the pilot, Table 4), `analyze_iappi.py` (label-free IA-PPI code check on the pilot), and `collect_free.py`—a multi-provider free-tier collector (21 judge configs / 13 families over Groq, Google, Cerebras, Mistral, OpenRouter, Cohere, SambaNova, NVIDIA, GitHub Models, Hugging Face, Together; Anthropic optional). Pre-registration and code: [https://github.com/\[REDACTED\]](https://github.com/[REDACTED]), [https://osf.io/\[REDACTED\]](https://osf.io/[REDACTED]).

Broader Impact

Bias correction shapes model rankings and RLHF reward labels, so both under- and over-correction matter. We flag that the factorial rectifier with $N_{\text{cal}} < 500$ can *worsen* bias, and caution against citing our simulated magnitudes as properties of any named model.

References

- [1] L. Zheng et al., “Judging LLM-as-a-judge with MT-Bench and Chatbot Arena,” in *NeurIPS*, 2023.
- [2] J. Gu et al., “A survey on LLM-as-a-judge,” arXiv:2411.15594, 2024.

- [3] D. Li et al., “From generation to judgment: Opportunities and challenges of LLM-as-a-judge,” arXiv:2411.16594, 2024.
- [4] L. Shi et al., “Judging the judges: A systematic study of position bias in LLM-as-a-judge,” in *IJCNLP-AACL*, 2025. arXiv:2406.07791.
- [5] P. Wang et al., “Large language models are not fair evaluators,” in *ACL*, 2024. arXiv:2305.17926.
- [6] K. Saito et al., “Verbosity bias in preference labeling by large language models,” arXiv:2310.10076, 2023.
- [7] A. Panickssery et al., “LLM evaluators recognize and favor their own generations,” arXiv:2404.13076, 2024.
- [8] X. Wu et al., “Self-preference bias in LLM-as-a-judge,” arXiv:2410.21819, 2024.
- [9] Y. Dubois et al., “Length-controlled AlpacaEval: A simple way to debias automatic evaluators,” arXiv:2404.04475, 2024.
- [10] J. Ye et al., “Justice or prejudice? Quantifying biases in LLM-as-a-judge,” arXiv:2410.02736, 2024.
- [11] A. S. Thakur et al., “Judging the judges: Evaluating alignment and vulnerabilities in LLMs-as-judges,” arXiv:2406.12624, 2024.
- [12] S. K. Soumik, “Judging the judges: A systematic evaluation of bias mitigation strategies in LLM-as-a-judge pipelines,” *Trans. Mach. Learn. Res.*, 2026. arXiv:2604.23178.
- [13] “Any large language model can be a reliable judge: Debiasing with a reasoning-based bias detector,” arXiv:2505.17100, 2025.
- [14] R. Koo et al., “Benchmarking cognitive biases in large language models as evaluators,” arXiv:2309.17012, 2023.
- [15] M. Sharma et al., “Towards understanding sycophancy in language models,” arXiv:2310.13548, 2023.
- [16] M. Wu and A. F. Aji, “Style over substance: Evaluation biases for large language models,” arXiv:2307.03025, 2023.
- [17] A. N. Angelopoulos et al., “Prediction-powered inference,” *Science*, vol. 382, pp. 669–674, 2023.
- [18] A. N. Angelopoulos, J. C. Duchi, and T. Zrnic, “PPI++: Efficient prediction-powered inference,” arXiv:2311.01453, 2023.
- [19] A. Divekar, “Statistically reliable LLM-based ranking evaluation via prediction-powered inference,” in *ACL 2026 GEM Workshop*. arXiv:2606.05308.
- [20] Y. T. Chen, S. Lu, S. Li, M. Guo, and S. Li, “Efficient inference for noisy LLM-as-a-judge evaluation,” arXiv:2601.05420, 2026.
- [21] R. Brant, “Assessing proportionality in the proportional odds model,” *Biometrics*, vol. 46, pp. 1171–1178, 1990.
- [22] J. D. Angrist and J.-S. Pischke, *Mostly Harmless Econometrics*. Princeton Univ. Press, 2009.
- [23] C. Guo et al., “On calibration of modern neural networks,” in *ICML*, 2017.

A Detection Protocol

The interaction-detection endpoint and its decision rule were fixed in advance of the pilot: primary endpoint $\hat{i}_{PV}$ (the probability-scale swap-cell DiD) per judge and pooled; decision rule “ $\hat{i}_{PV}$ 95% CI excludes 0 and $|\hat{i}_{PV}| > 0.02 \Rightarrow$ sequential correction inadequate.” SSIT is the estimator for that endpoint.

Target design for the full study.

- ≥ 8 judges over ≥ 5 model families, *including* frontier deployed judges (GPT-4-class, Claude, Gemini) alongside open models.
- Stimuli quality-matched by an independent rater ($|\Delta\text{quality}| < 0.1$); ≥ 50 pairs $\times$ 4 task types.
- Verbosity manipulated by *padding* r_1 with redundant but on-topic sentences (length only), not paraphrase expansion, to unconfound length from rephrasing.
- All eight (P, V, S) swap cells, ≥ 3 samples/cell; self-preference cells use a held-out generator, not the judge’s own output, to unconfound S from length/style.
- Inference: wild-cluster-bootstrap CI (robust with few clusters); Bonferroni over the three factor pairs.

The reported pilot is a small initial slice that omits the frontier judges, the padding manipulation, and the multiplicity control.

IASC and IA-PPI were developed after the initial two-judge collection and are corrections evaluated *post hoc* on the same frozen data—not part of the pre-registered analysis. We flag them as such throughout.

B Proof of Proposition 2

Under $\text{logit}\pi_1(P, V) = \delta + \beta_P(2P-1) + \beta_V V + \beta_{PV}(2P-1)V$: $\text{logit}\pi_1(1, 1) - \text{logit}\pi_1(0, 1) = 2\beta_P + 2\beta_{PV}$ and $\text{logit}\pi_1(1, 0) - \text{logit}\pi_1(0, 0) = 2\beta_P$; their difference is $2\beta_{PV}$. Every term in δ and β_V (which do not depend on P) cancels in the inner differences, and β_P cancels in the outer difference. $\square$

Proposition 3 (IASC shrinkage). The four cell log-odds identify β_P (inner difference = $2\beta_P$) and β_{PV} (DiD = $2\beta_{PV}$) free of δ, β_V . The plug-in fits a *single* slope pair across pairs whose δ_k vary; a pooled logistic slope under an omitted heterogeneous intercept attenuates, $\hat{\beta}_\bullet \xrightarrow{P} \lambda_\bullet \beta_\bullet$ with $\lambda_\bullet \in (0, 1]$ and $\lambda_\bullet \rightarrow 1$ as $\text{Var } \delta \rightarrow 0$ (mixed-logit / Jensen). Substituting into (4), $\mathbb{E}[\text{logit}\hat{s}_i | k] = \delta_k + (1-\lambda_P)\beta_P(2P_i-1) + \beta_V V_i + \beta_S S_i + (1-\lambda_{PV})\beta_{PV}(2P_i-1)V_i$: a positive fraction of each position-coupled term is removed, sign preserved. λ_P and λ_{PV} need not be equal, so a cell can retain oppositely-signed main-effect and interaction residuals; and the shrinkage guarantee is on the logit scale—after σ the Jensen gap can push the corrected pick rate outside [raw, swap-avg]. At $\lambda_P = \lambda_{PV} = 1$ (matched calibration pairs) IASC coincides with $2\times$ swap-averaging in the log-odds. $\square$

C Optional CI Companion: the Factorial Rectifier

This appendix is a companion, not a core contribution: it says how to attach a *valid confidence interval* to a debiased win rate when a small gold set *is* available. Reference implementation: `ia_ppi.py`.

Prediction-powered inference (PPI++ [18, 20]) debiases a large judge-labelled set $\mathcal{E}$ against a small gold set $\mathcal{C}$ using a rectifier $\mu(\hat{y}) = \mathbb{E}[H | \hat{Y}=\hat{y}]$.

Proposition 4 (Marginal PPI under-covers under cell shift). *With $g(\hat{y}, c) = \mathbb{E}[H | \hat{Y}=\hat{y}, c] - \mu(\hat{y})$ and π_c, π'_c the cell frequencies in $\mathcal{C}, \mathcal{E}$, $\hat{\theta}_{\text{PPI}}$ has asymptotic bias $\sum_c (\pi'_c - \pi_c) \mathbb{E}[g | c]$ with $\mathbb{E}[g | c] = \frac{1}{4} s_c \beta_{PV} + O(\beta_{PV}^2)$; the coverage loss is order (bias/SE). The factorial rectifier $\mu(\hat{y}, P, V, PV)$ —fit on $\mathcal{C}$ or seeded from SSIT—sets $g \equiv 0$; the resulting estimator is $\sqrt{N_{\text{cal}}}$ -asymptotically normal with a sandwich variance, and its point estimate equals the joint $P+V+PV$ plug-in. Proof: standard PPI augmentation with a cell-stratified rectifier (`ia_ppi.py`, `docstring`).*

Simulation (`sim_prop2_coverage.py`, $N_{\text{cal}}=800, 1,500$ reps, $\mathcal{E}$ on the joint-bias cell): marginal-PPI 95% coverage falls to ≈ 0.89 as the interaction grows and to 0.82 under a probit judge link; the factorial rectifier holds 0.94–0.95 throughout, and also under three variants where its working model is wrong (probit link; an active three-way term; a quality-dependent position slope), so it is not merely recovering its own DGP. Below $N_{\text{cal}} \approx 500$ the factorial rectifier is too noisy and increases residual bias relative to sequential correction, so it needs a few hundred gold pairs to be worth using.

D Pilot: Dropped Verdicts

$\approx 15\%$ of gemma responses, $\approx 9\%$ of qwen3-27b, and $< 3\%$ of gpt-oss did not contain a parseable VERDICT: line and were dropped (allam-2-7b, mistral-small, command-r: 0%). The drop rate is mildly cell-correlated (highest in the expanded- V cells, where gemma and qwen3-27b more often deliberate instead of committing), which could bias $\hat{i}_{PV}$; under the padding-only V qwen3-27b’s drop rate rises to $\approx 16\%$ (redundant restatement makes it more often call the two responses equivalent), a caveat on its null there. The wild cluster bootstrap in Table 3 treats a pair with a missing cell as absent rather than imputing. The pre-registered protocol (Appendix A) fixes ≥ 3 samples/cell so a single unparseable draw does not empty a cell, and logs the raw generations for post-hoc coding.

E IASC on the Pilot

Table 4: IASC on the pilot (`analyze_iasc.py`). $\hat{\beta}_P, \hat{\beta}_{PV}$ are the gold-free swap-cell estimates (log-odds); “raw” is the uncorrected target-pick rate on the single-order ($P=1$) slice, “+IASC” the same slice after Eq. (4) (pair-cluster bootstrap 95% CI, coefficients re-fit per resample), “swap-avg.” the full $2 \times$ swap-averaged rate. IASC and swap-averaging move the raw estimate the same direction; they differ by the logit-vs-probability Jensen gap.

Judge	$\hat{\beta}_P$	$\hat{\beta}_{PV}$	raw	+IASC	swap-avg.
gpt-oss-20b	−1.13	+1.77	0.96	0.98 [0.97, 1.00]	0.97
gpt-oss-120b	−1.48	+1.48	0.86	0.94 [0.92, 0.98]	0.93
gemma	−1.58	+1.43	0.87	0.94 [0.92, 0.98]	0.89
qwen3-27b	−1.53	+2.27	0.95	0.99 [0.98, 0.99]	0.97
allam-2-7b	−1.60	+0.77	0.89	0.97 [0.96, 0.98]	0.93
mistral-small	−0.52	+0.52	0.94	0.96 [0.94, 0.98]	0.96
command-r	0.00	0.00	1.00	1.00	1.00
Pooled	−1.36	+1.28	0.93	0.98 [0.97, 0.99]	0.95

F All Three Factor Pairs and Task Breakdown on the Pilot

Gold-free SSIT applied to each factor pair (`analyze_real.py`; probability-scale DiD $\hat{i}_{XY}$, pair-cluster bootstrap 95% CI over ≈ 20 clusters, $B=3,000$). “AIA rej.” = the $\hat{i}_{XY}$ CI and the log-odds $\hat{b}_{XY}$ CI both exclude 0 (nominal, *not* multiplicity-corrected; three pairs $\times$ four groupings).

Judge	Pair	$\hat{i}_{XY}$	AIA rej.
Pooled (7)	$P \times V$	+0.10 [+0.05, +0.14]	yes
Pooled (7)	$P \times S$	+0.08 [+0.03, +0.15]	no
Pooled (7)	$V \times S$	-0.06 [-0.11, -0.00]	no
gpt-oss-20b	$P \times V$	+0.12 [+0.03, +0.23]	yes
gpt-oss-120b	$P \times V$	+0.17 [+0.06, +0.31]	yes
gemma	$P \times V$	+0.16 [+0.08, +0.25]	yes
qwen3-27b	$P \times V$	+0.12 [+0.05, +0.20]	yes
allam-2-7b	$P \times V$	+0.09 [+0.00, +0.16]	no
mistral-small	$P \times V$	+0.07 [+0.01, +0.16]	no
command-r	$P \times V$	+0.00 [+0.00, +0.00]	no
gpt-oss-120b	$V \times S$	-0.17 [-0.31, -0.06]	yes
mistral-small	$V \times S$	-0.12 [-0.25, -0.03]	yes
qwen3-27b	$V \times S$	-0.09 [-0.18, -0.00]	no
allam-2-7b	$V \times S$	-0.09 [-0.16, -0.01]	no
gemma	$V \times S$	+0.11 [-0.01, +0.24]	no

$P \times V$ is positive on six of seven judges (all but command-r, which is not challenged by the stimulus), pooled +0.10 (+0.12 excluding command-r). $P \times S$ points the same way pooled but its $\hat{i}$ and log-odds CIs do not both clear 0, and the $S=1$ cells substitute the judge’s own generation for r_1 , so it is confounded with $P \times (\text{length/style})$ —we do not claim it. $V \times S$ (outside the swap subspace) is negative for four judges, positive for one, weakly negative pooled (-0.06)—the selective pattern Prop. 3 predicts, but on seven judges $\times$ 20 pairs it is suggestive, not established.

By task (pooled over seven judges, $\hat{i}_{PV}$): Math Reasoning +0.23 [+0.19, +0.28]; Summarization +0.06 [+0.02, +0.09] (AIA rej.); Open-ended QA +0.02 [+0.00, +0.04]; Code Review omitted (3 pairs). The interaction concentrates on Math Reasoning and never sign-reverses.

G Leaderboard-Impact Sketch

A *constructed* demo (sim_ranking_impact.py): six models on a tight skill ladder, judge carrying $\hat{i}_{PV} = +0.15$, verbosity aligned adversarially with rank. Real leaderboards randomise position, so the absolute τ -drop is an artefact; the ordering of methods is the point. Sweeping $\beta_{PV} = -\beta_P$, Kendall τ to the true order: no correction and sequential correction stay broken ($\tau = +0.20$) at every magnitude; log-odds swap-averaging and joint $P+V+PV$ estimation stay exact; label-free IASC ranges $\tau \in [+0.60, +0.87]$, degrading with the bias magnitude as λ_{PV} falls. Details and the full table in the code.